



MASENO UNIVERSITY

UNIVERSITY EXAMINATIONS 2016/2017

**FIRST YEAR SECOND SEMESTER EXAMINATIONS FOR THE
DEGREE OF BACHELOR OF EDUCATION, BACHELOR OF
SCIENCE AND BACHELOR OF ARTS WITH INFORMATION
TECHNOLOGY**

CITY CAMPUS

MMA 101: ANALYTICAL GEOMETRY

Date: 1st October, 2017

Time: 8.30 - 11.30 am

INSTRUCTIONS:

- Answer question ONE and any other TWO questions
- Electronic scientific calculators may be used
- Observe further instructions on the answer booklet



QUESTION ONE (Compulsory)

[30 Marks]

- (a) State the eccentricity of the following conic sections: [3 Marks]
- (i) Parabola
 - (ii) Ellipse
 - (iii) Hyperbola
- (b) Determine the equation of a straight line whose slope is 4 and the y - intercept is 3. [2 Marks]
- (c) Given two lines $L_1 : y = 3x + 4$ and $L_2 : 4y + 3x + 1 = 0$. Find;
- (i) the point of intersection of the two lines [2 Marks]
 - (ii) the angle between the two lines [3 Marks]
 - (iii) the distance from the point $(1, -3)$ to L_2 . [3 Marks]
- (d) Determine the radius and the coordinates of the centre of a circle given by the equation $2x^2 + 2y^2 - 8x + 5y + 10 = 0$. [4 Marks]
- (e) Identify the conics given below: [4 Marks]
- (i) $x^2 - 4xy + 4y^2 - 24x + 12y = 0$
 - (ii) $x^2 - 2xy + 2y^2 - 25 = 0$
 - (iii) $xy = 2$
 - (iv) $4x^2 - y^2 - 4y + 8 = 0$.
- (f) Convert the following cartesian coordinates into polar coordinates: [4 Marks]
- (i) $(-4, 5)$
 - (ii) $(-2, -7)$.
- (f) A straight line passes through points $(-3, 4)$ and $(5, -6)$.
- (i) Write the equation of this line in double intercept form. [3 Marks]
 - (ii) Find the equations of a line perpendicular to this line and passes through point $(1, 3)$. [2 Marks]

QUESTION TWO

[20 Marks]

- (a) Given two parallel lines $L_1 : 4y - 3x = -3$ and $L_2 : 4y = 3x - 28$. Determine the distance between the lines L_1 and L_2 and the equation of a line that passes midway between them. [6 Marks]
- (b) Find the equation of a circle which passes through the points $(6, -5)$, $(2, -7)$ and $(-6, -1)$ giving your answer in the form $Ax^2 + By^2 + Cx + Dy + E = 0$. Determine the area of the circle. [10 Marks]
- (c) Give the cartesian coordinates of the following polar coordinates: [4 Marks]
- (i) $(5, 70^\circ)$
 - (ii) $(-3, -300^\circ)$.

QUESTION THREE

[20 Marks]

The equation of an ellipse on the xy -plane is given by $9x^2 + 4y^2 + 36x - 8y + 4 = 0$. Find

- (a) the centre of the ellipse [5 Marks]
- (b) the coordinates of the vertices [4 Marks]
- (c) the foci [3 Marks]
- (d) the eccentricity [2 Marks]
- (e) the directrices [3 Marks]
- (f) the area of the ellipse [3 Marks]

QUESTION FOUR

[20 Marks]

- (a) Consider the equation of a conic section given by $9y^2 - 16x^2 - 144 = 0$.
- (i) Show that the equation represent a hyperbola. [2 Marks]
 - (ii) Determine
 - * the centre [1 Mark]
 - * the foci [3 Marks]
 - * the vertices [2 Marks]
 - * the directrices [2 Marks]
 - * the eccentricity [2 Marks]
 - * the asymptotes. [4 Marks]
- (b) Find the parametric representation of the following cartesian equations: [4 Marks]
- (i) $2x - 1 = 3y$
 - (ii) $x^2 + y^2 = 4$.

QUESTION FIVE

[20 Marks]

- (a) Sketch the polar curves of each of the following in the range $0^\circ \leq \theta \leq 360^\circ$ and give the name of each; [10 Marks]

(i) $r = 2 + 2 \cos \theta$

(ii) $r = 2 + 4 \cos \theta$

- (b) A parabola has the y -intercepts -1 and 2 . Find [10 Marks]

(i) the equation of the parabola

(ii) the equation of the axis of the parabola

(iii) the vertex and focus of the parabola

(iv) the equation of the directrix of the parabola



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INFORMATION TECHNOLOGY**

CITY CAMPUS

MMA 101: ANALYTICAL GEOMETRY

Date: 27th April 2018

Time 2.30 - 5.30pm

INSTRUCTIONS:

- Answer Question ONE (30Marks) and any other TWO questions (20 Marks each).
- Electronic Scientific calculators may be used.
- Observe further instructions in the answer booklet.



QUESTION ONE (Compulsory)

[30 Marks]

(a) State the eccentricity of the following conic sections: [3 Marks]

- (i) Parabola
- (ii) Ellipse
- (iii) Hyperbola

(b) Determine the equation of a straight line whose slope is 4 and the y - intercept is 3. [2 Marks]

(c) Given two lines $L_1 : y = 3x + 4$ and $L_2 : 4y + 3x + 1 = 0$. Find:

- (i) the point of intersection of the two lines [2 Marks]
- (ii) the angle between the two lines [3 Marks]
- (iii) the distance from the point $(1, -3)$ to L_2 . [3 Marks]

(d) Determine the radius and the coordinates of the centre of a circle given by the equation $2x^2 + 2y^2 - 8x + 5y + 10 = 0$. [4 Marks]

(e) Identify the conics given below: [4 Marks]

- (i) $x^2 - 4xy + 4y^2 - 24x + 12y = 0$
- (ii) $x^2 - 2xy + 2y^2 - 25 = 0$
- (iii) $xy = 2$
- (iv) $4x^2 - y^2 - 4y + 8 = 0$.

(f) Convert the following cartesian coordinates into polar coordinates: [4 Marks]

- (i) $(-4, 5)$
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- (i) Write the equation of this line in double intercept form. [3 Marks]
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[20 Marks]

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- (f) the area of the ellipse [3 Marks]

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- (ii) Determine
- * the centre [1 Mark]
 - * the foci [3 Marks]
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 - * the directrices [2 Marks]
 - * the eccentricity [2 Marks]
 - * the asymptotes. [4 Marks]
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- (i) $2x - 1 = 3y$
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[20 Marks]

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(i) $r = 2 + 2 \cos \theta$

(ii) $r = 2 + 4 \cos \theta$

- (b) A parabola has the y -intercepts -1 and 2 . Find

[10 Marks]

- (i) the equation of the parabola
- (ii) the equation of the axis of the parabola
- (iii) the vertex and focus of the parabola
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TECHNOLOGY**

CITY CAMPUS

MMA 102: CALCULUS I

Date: 30th September, 2017

Time: 3.30 - 6.30 pm

INSTRUCTIONS:

- Answer question ONE and any other TWO questions
- Electronic scientific calculators may be used
- Observe further instructions on the answer booklet



QUESTION ONE (Compulsory)

[30 Marks]

- (a) (i) State the Intermediate Value Theorem. [1 mk]
(ii) Using the theorem stated above, show that the polynomial function $f(x) = x^3 - 2x - 5$ has a real root between 3 and 5. [3 mks]
- (b) Let f, g be real - valued functions given by $f(x) = \log x$, and $g(x) = \frac{1}{1-x^2}$. Determine the composite functions $(f \circ g)(x)$, $(g \circ f)(x)$ and hence or otherwise, evaluate $(g \circ f)(1/e)$. [3 mks]
- (c) Evaluate the following limits;
- (i) $\lim_{t \rightarrow 2} \frac{t^3 - 8}{t^2 - 4}$ [2 mks]
(ii) $\lim_{x \rightarrow \infty} \frac{x}{\sqrt{x^2 + 1}}$ [2 mks]
(iii) $\lim_{x \rightarrow 4} \frac{\sqrt{x} - 2}{x^2 - 5x + 4}$ [2 mks]
- (d) Using the definition of continuity, discuss the continuity of the function

$$f(x) = \begin{cases} 1 - 2x, & x < 0; \\ 2 - x, & x \geq 0, \end{cases}$$

at the point $x = 0$.

[2 mks]

- (e) Find the derivative of the following functions from first principles:

- (i) $f(x) = 3x^2$ [2 mks]
(ii) $f(x) = \cos x$ [2 mks]
- (f) Determine the slope of the tangent to the curve $f(x) = \sqrt[3]{x}$ at the point $x = 8$; hence write the equation of this tangent line. [3 mks]
- (g) Suppose that $f : \mathbb{R} \rightarrow \mathbb{R}$ is a differentiable function with $f(3) = 2$ and $3 \leq f'(x) \leq 4$ for all x . Show that $8 \leq f(5) \leq 10$. [3 mks]
- (h) (i) If a point traces the circle $2x^2 + 3y^2 = 25$ and if $\frac{dx}{dt} = 4$ when the point reaches $(3, 4)$, find $\frac{dy}{dt}$ at this point. [2 mks]
(ii) A body moves in a straight line according to the law of motion $S = 3t^3 + 4t^2 - 3t + 10$. Find its acceleration at each instant when the velocity is 0. [3 mks]

QUESTION TWO

[20 Marks]

Find $\frac{dy}{dx}$ in the following functions

(a) $y^2 = \cos(\sin \sqrt{x^2 + 1})$ [4 mks]

(b) $x^2 = \frac{x-y}{x+y}$ [3 mks]

(c) $y = \cos^{-1} 2x$ [3 mks]

(d) $y = x^3 \ln 2x$ [3 mks]

(e) $y = x^{x^x}$ [4 mks]

(f) $y = \frac{(x^2+9)^2(x-3)^4}{x^2+2}$ [3 mks]

QUESTION THREE

[20 Marks]

(a) By differentiating $x^2 - y^2 = 1$ implicitly, show that

$$y'' = -\frac{1}{y^3}$$

[5 mks]

(b) Write the polynomial

$$f(x) = -35 + 45x - 20x^2 + 3x^3$$

in the form

$$f(x) = C_0 + C_1(x-3) + C_2(x-3)^2 + C_3(x-3)^3$$

where the C_i 's ($i = 0, 1, 2, 3$) are constants.

[6 mks]

(c) Consider the curve $y = x^5 - 5x^3$

(i) Classify all the stationary points of the curve, and hence

[5 mks]

(ii) sketch the curve

[4 mks]

QUESTION FOUR

[20 Marks]

- (a) Find the slope of the tangent to the curve $x^2 + xy + y^2 = 7$ at the point $(1, 2)$. Hence, find the equation of the normal to the curve at this point. [6 mks]
- (b) For oscillations of small amplitude, the relation between the period T and the length L of a simple pendulum may be approximated by the equation

$$T = 2\pi\sqrt{\frac{L}{g}},$$

where $g = 980\text{cm s}^{-2}$ is the acceleration due to gravity. When the temperature θ changes, the length L either increases or decreases at a rate that is proportional to L , that is

$$\frac{dL}{L} \propto L.$$

Show that the rate of change of period with respect to temperature is given by

$$\frac{dT}{d\theta} = \frac{kT}{2}$$

where k is constant of proportionality. [4 mks]

- (c) Find the gradient of the curve $x = \frac{t}{1+t}$, $y = \frac{t^3}{1+t}$ at the point $(\frac{1}{2}, \frac{1}{2})$. [6 mks]
- (d) Use L' Hopital's rule or otherwise to evaluate the following limits;
- (i) $\lim_{x \rightarrow 0} x \ln x$ [2 mks]
- (ii) $\lim_{x \rightarrow 0^+} x^x$ [2 mks]

QUESTION FIVE

[20 Marks]

- (a) 64cm^3 of butter is to be made into a slab of square cross - section. Compute the required lengths if the total surface area is to be as small as possible. [5 mks]
- (b) (i) State the Mean Value Theorem [2 mks]
- (ii) Use the Mean Value Theorem to prove that $|\sin a - \sin b| \leq |a - b|$ for all real values of a and b where $a \neq b$. [4 mks]
- (c) A car passes a camera at a point A on the toll road with speed 50km/h . One hour later the same car passes a camera at a point B , located 100km down the road from camera A , traveling at 50km/h . Prove that the car was overspeeding at some point along the road if the designated speed limit is 80km/h . [5 mks]
- (d) Find the points on the curve $y = 2x^3 - 3x^2 - 12x + 20$ where the tangent is parallel to the x -axis. [4 mks]

END
ALL THE BEST



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**SECOND YEAR SECOND SEMESTER EXAMINATION
FOR THE DEGREE OF BACHELOR OF EDUCATION WITH
INFORMATION TECHNOLOGY**

CITY CAMPUS

MMA 201: LINEAR ALGEBRA II

Date: 29th April, 2018

Time: 8.30 - 11.30am

INSTRUCTIONS:

- Answer Question ONE and any other TWO
- Proofs should be written carefully
- Cheating is NOT allowed and will be harshly punished
- Observe further instructions on the answer booklet



QUESTION ONE (Compulsory)

[30 Marks]

- (a) (i) Define a vector subspace [2 mks]

- (ii) Show that the set $W = \left\{ \begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2 : x \geq 0, y \geq 0 \right\}$ is NOT a subspace of $\mathbb{R}^2$.
[3 mks]

- (b) Let P_2 denote the space of all polynomials $f(t)$ of degree ≤ 2 and V be its subspace such that $f'(1) = 0$. Find the basis and dimension of V . [3 mks]

- (c) Determine whether the matrices $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$, $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $\begin{bmatrix} 2 & 3 \\ 5 & 7 \end{bmatrix}$, and $\begin{bmatrix} 1 & 4 \\ 6 & 8 \end{bmatrix}$ are linearly independent. [5 mks]

- (d) Let $C[0, 1]$ be the linear space of all continuous functions $f : [0, 1] \rightarrow \mathbb{R}$. Define $I : C[0, 1] \rightarrow \mathbb{R}$ by

$$I(f) = \int_0^1 f(t) dt.$$

- (i) Show that I is linear [3 mks]

- (ii) Find the image of I [1 mk]

- (e) Suppose that a linear mapping $T : U \rightarrow V$ between vector spaces U and V is an isomorphism.

- (i) Explain the meaning of an isomorphism. [2 mks]

- (ii) What is the image of T , $Im(T)$, and the kernel of T , $ker(T)$. [2 mks]

- (iii) If $\{f_1, f_2, \dots, f_n\}$ is a basis of U , determine the basis of V . [2 mks]

- (iv) If T is not an isomorphism, show that $Im(T)$ is a subspace of V . [3 mks]

- (f) Consider the ordered bases $\beta = \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}$ and $\alpha = \left\{ \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 2 \\ 1 \end{pmatrix} \right\}$ for the vector space $\mathbb{R}^2$. Compute ${}_{{\alpha}}P_{\beta}$, the transition matrix from β to α . [4 mks]

QUESTION TWO

[20 Marks]

- (a) Consider the linear space V that consists of all functions of the form $f(t) = c_1 \cos t + c_2 \sin t$ where c_1 and c_2 are arbitrary constants.
- (i) Find the matrix of the linear transformation $T : V \rightarrow V$ given by $T(f) = f'' + af' + bf$ with respect to the basis $\{\cos t, \sin t\}$ where a and b are arbitrary constants. [5 mks]
- (ii) Give a condition for the isomorphism of T . [3 mks]
- (iii) Find the functions f in V such that [5 mks]

$$T(f) = f'' + af' + bf = \cos t.$$

- (b) Evaluate the determinant of the following matrix [7 mks]

$$\begin{pmatrix} -5 & 1 & 1 & -6 & 2 \\ 0 & 1 & 3 & 8 & -4 \\ -1 & 2 & 1 & 8 & 9 \\ 0 & 0 & 1 & 14 & 2 \\ 1 & 1 & 0 & 0 & 0 \end{pmatrix}$$

QUESTION THREE

[20 Marks]

- (a) Determine whether or not the matrix

$$H = \begin{bmatrix} 15 & 8i & 6 - 2i \\ -8i & 0 & -4 + i \\ 6 + 2i & -4 - i & -3 \end{bmatrix}$$

is Hermitian.

[5 mks]

- (b) Let $Z = \begin{bmatrix} z_1 \\ z_2 \\ \vdots \\ z_n \end{bmatrix}$ be any complex $n \times 1$ matrix. Show that if S is skew-hermitian, then $\bar{Z}^T S Z$ is either zero or pure imaginary. [5 mks]

- (c) Let S denote the space of all skew-symmetric 4×4 matrices.

- (i) Find the basis of S . [7 mks]
- (ii) Determine the dimension of S . [3 mks]

QUESTION FOUR

[20 Marks]

- (a) Consider the matrix A given by

$$\begin{pmatrix} 5 & 0 & 1 \\ 0 & 1 & 2 \\ 1 & 0 & 3 \end{pmatrix}.$$

- (i) Find the eigenvalues of A . [5 mks]
 - (ii) Find the eigenvectors associated with the eigenvalues obtained in (i) above. [6 mks]
 - (iii) Diagonalize the matrix A . [4 mks]
- (b) Evaluate the Vandermondes determinant

$$\begin{vmatrix} 1 & \alpha & \alpha^2 \\ 1 & \beta & \beta^2 \\ 1 & \gamma & \gamma^2 \end{vmatrix}$$

where α , β and γ are scalars.

[5 mks]

QUESTION FIVE

[20 Marks]

- (a) Give any two properties of eigenvalues and eigenvectors of a real symmetric matrix. [4 mks]
- (b) State and prove the Cayley Hamilton theorem. [5 mks]
- (c) Consider the matrix A given by

$$A = \begin{bmatrix} 3 & 0 & -2 \\ 0 & 2 & 0 \\ -2 & 0 & 0 \end{bmatrix}.$$

- (i) Show that the matrix A satisfies its own characteristic equation. [6 mks]
- (ii) Use Cayley-Hamilton theorem to compute A^{-1} , the inverse of A . [5 mks]

END
ALL THE BEST



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SECOND YEAR SECOND SEMESTER EXAMINATIONS FOR THE DEGREE OF BACHELOR OF EDUCATION WITH INFORMATION TECHNOLOGY

CITY CAMPUS

MMA 202: VECTOR ANALYSIS

Date: 30th September, 2017

Time: 12.00 - 3.00 pm

INSTRUCTIONS:

- Answer question ONE and any other TWO questions
- Proofs should be written carefully
- Cheating is NOT allowed and will be harshly punished
- Observe further instructions on the answer booklet



QUESTION ONE (Compulsory)

[30 Marks]

- (a) Differentiate between a vector and a scalar, giving an example in each case. [4 Marks]
- (b) The forces F_1 , and F_2 act on an object O and the object moves. State the forces that are needed to prevent O from moving? [2 Marks]
- (c) Show that addition of vectors is commutative. [3 Marks]
- (d) Show that if $\mathbf{u}$ and $\mathbf{v}$ are non collinear, then $x\mathbf{u} + y\mathbf{v} = 0$ implies that $x = 0$ and $y = 0$. [3 Marks]
- (e) (i) Prove that $\nabla \cdot (\nabla \times \mathbf{A}) = 0$. [3 Marks]
- (ii) If $\mathbf{A} = (xy^2, \sin y, z^2)$, find $\text{curl } \mathbf{A}$ at $(1, 3, -1)$. [3 Marks]
- (f) The acceleration of any particle at any time t is given by

$$\mathbf{a} = 12 \cos t \mathbf{i} - 8 \sin 2t \mathbf{j} + 16t \mathbf{k}.$$

If the velocity $\mathbf{V}$ and displacement $\mathbf{r}$ are 0 at $t = 0$. Find $\mathbf{V}$ and $\mathbf{r}$ at any time.

[4 Marks]

- (g) Evaluate the integral

$$\int_C [ydx - xdy]$$

using Green's theorem, where C is the circumference of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

[4 Marks]

- (h) Use divergence theorem to find the flux output from the ellipsoid of volume V if $\mathbf{F} = 3x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$. [4 Marks]

QUESTION TWO

[20 Marks]

- (a) Given the space curve $x = t$, $y = t^2$, $z = \frac{2}{3}t^3$. Find the:
- (i) curvature, κ , and its radius, ρ . [8 Marks]
- (ii) torsion, τ , and its radius, σ . [6 Marks]
- (b) Find the directional derivative of $f(x, y, z) = x^2yz + 4xz^2$ at $(1, -2, -1)$ in the direction of $2\mathbf{i} - \mathbf{j} - 2\mathbf{k}$. [6 Marks]

QUESTION THREE

[20 Marks]

- (a) If $\mathbf{A} = (3x^2 + 6y)\mathbf{i} - 14yz\mathbf{j} + 20xz^2\mathbf{k}$. Evaluate $\int_C \mathbf{A} \cdot d\mathbf{R}$ from $(0, 0, 0)$ to $(1, 1, 1)$ along the following curves C .

(i) $x = t, y = t^2, z = t^3$. [3 Marks]

- (ii) the straight lines from $(0, 0, 0)$ to $(1, 0, 0)$ then to $(1, 1, 0)$ and finally to $(1, 1, 1)$. [4 Marks]

- (iii) the straight line joining $(0, 0, 0)$ to $(1, 1, 1)$. [3 Marks]

- (b) (i) State Green's theorem. [2 Marks]

- (ii) Evaluate the integral along the curve C .

$$\int_C [(y - x^2 e^x)dx + (\cos 2y^2 - x)dy]$$

where C is the rectangle with vertices $(1, 1)$, $(0, 1)$, $(1, 3)$ and $(0, 3)$. [8 Marks]

QUESTION FOUR

[20 Marks]

- (a) Determine the constant c such that the vector $\mathbf{V} = (x + 3y)\mathbf{i} + (y - 2z)\mathbf{j} + (x + cz)\mathbf{k}$ is solenoidal. [5 Marks]

- (b) (i) State Stoke's theorem. [2 Marks]

- (ii) Verify Stoke's theorem for $\mathbf{A} = (2x - y)\mathbf{i} - yz^2\mathbf{j} - y^2zk$ where S is the upper half surface of the sphere $x^2 + y^2 + z^2 = 1$ and C is its boundary. [13 Marks]

QUESTION FIVE

[20 Marks]

- (a) Evaluate

$$\iint_S \mathbf{A} \cdot \mathbf{n} dS$$

where $\mathbf{A} = 18z\mathbf{i} - 12\mathbf{j} + 3y\mathbf{k}$ and S is that part of the plane $2x + 3y + 6z = 12$ which is located in the first octant. [10 Marks]

- (b) (i) State divergence theorem. [2 Marks]

- (ii) Use divergence theorem to evaluate

$$\iint_S \mathbf{A} \cdot \mathbf{n} dS$$

where $\mathbf{A} = 4xz\mathbf{i} - y^2\mathbf{j} + yz\mathbf{k}$ and S is the surface of the cube bounded by $x = 0, x = 1, y = 0, y = 1, z = 0, z = 1$. [8 Marks]



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**SECOND YEAR SECOND SEMESTER EXAMINATIONS FOR
THE DEGREE OF BACHELOR OF EDUCATION SCIENCE
WITH INFORMATION TECHNOLOGY**

CITY CAMPUS

**MMA 211: PROBABILITY AND DISTRIBUTION
THEORY II**

Date: 1st October, 2017

Time: 8.30 - 11.30 am

INSTRUCTIONS:

- Answer question ONE and any other TWO questions
- Show all the necessary workings
- Begin each question on a new page



QUESTION ONE (COMPULSORY) (30mks)

(a) Define the following terms:

(i) Central limit theorem

(ii) Standard error of the mean [2mks]

(b) (i) The weekly wage of 2000 workmen is normally distributed with mean wage of Rs 70 and wage standard deviation of Rs 5. Estimate the number of workers whose weekly wages are between Rs 69 and Rs 73 [4mks]

(ii) The amount of fuel consumed by the engines of a jetliner on a flight between two cities is a normally distributed random variable X with mean $\mu = 5.7$ tons and standard deviation $\sigma = 0.5$ tons. Carrying too much fuel is inefficient as it slows the plane. If, however, too little fuel is loaded on the plane, an emergency landing may be necessary. What should be the amount of fuel to load so that there is 0.99 probability that the plane will arrive at its destination without emergency landing? [5mks]

(c) A random variable X has mean 10 and variance 4 and unknown probability distribution. Using Chebyshev's theorem, find

(i) $p(|x - 10| \geq 3)$ [4mks]

(ii) the value of the constant c such that $p(|x - 10| \geq c) \leq 0.04$ [4mks]

(d) In an automated process, a machine fills cans of coffee. The variance of the filling process is known to be 30. In order to keep the process in control, from time to time regular checks of the variance of the filling process are made. This is done by randomly sampling filled cans, measuring their amounts and computing the sample variance. A random sample of 101 cans is selected for the purpose. What is the probability that the sample variance is between 21.28 and 38.72? [5mks]

(e) Let a discrete population be given by the numbers $\{1, 2, 3\}$. If all random samples of size 2 are drawn without replacement, obtain the following in each case

(i) The sampling distribution of the sample mean

(ii) The mean and variance of the sample mean. [6mks]

QUESTION TWO (20mks)

(a) (i) State and prove Chebyshev's Inequality

(ii) A random variable X has mean 24 and variance 9. Obtain a bound on the probability that the random variable X assumes values between 16.5 to 31.5 [10mks]

(b) Let the random variable x and y have a joint density function

$$f(x, y) = \begin{cases} \frac{1}{2}(x + y)e^{-(x+y)}, & x, y > 0 \\ 0, & \text{elsewhere} \end{cases}$$

Find the density function of $z = x + y$ [5mks]

(c) A fruit drink company wants to know the variation, as measured by the standard deviation, of the amount of juice in 16 ounce cans. From past experience, it is known that $\sigma^2 = 2$. The company statistician decides to take a sample of 25 cans from the population line and compute the sample variance. Assuming that the sample values may be viewed as a random sample from a normal population, find a value of b such that $p(S^2 > b) = 0.05$, (S^2 - sample variance) [5mks]

QUESTION THREE (20mks)

- (a) Let x_1 and x_2 be two random variables such that

$$f(x_1, x_2) = \begin{cases} 6e^{-3x_1-2x_2} & x_1 > 0, x_2 > 0 \\ 0 & \text{elsewhere} \end{cases}$$

Find the pdf of $Y = x_1 + x_2$ [7mks]

- (b) Let s_1^2 denote the sample variance for a random sample of size 10 from Population I and let s_2^2 denote the sample variance for a random sample of size 8 from Population II. The variance of Population I is assumed to be three times the variance of Population II. Find two numbers a and b such that $P(a \leq \frac{s_1^2}{s_2^2} \leq b) = 0.90$, assuming s_1^2 is independent of s_2^2 . [6mks]

- (c) Suppose that we are attempting to locate a target in three-dimensional space, and that the three coordinate errors (in meters) of the point chosen are independent normal random variables with mean 0 and standard deviation 2. Find the probability that the distance between the point chosen and the target exceeds 3 meters. [4mks]

- (d) A cost accountant needs to forecast the unit cost of a product for the next year. He notes that each unit of the product requires 10 labour hours and 5 kg of raw material. In addition, each unit of the product is assigned an overhead cost of sh. 200. He estimates that the cost of a labour hour next year will be normally distributed with an expected value of Sh. 45 and a standard deviation of sh. 2, the cost of raw material will be normally distributed with an expected value of sh. 60 and a standard deviation of sh. 3.

(i) Find the distribution of the unit cost of the product

(ii) Find its expected value and variance. [3mks]

QUESTION FOUR (20mks)

- (a) A manufacturer of screws has noticed that on average, 0.02 proportion of screws produced are defective. A random sample of 400 screws is examined for the proportion of defective screws. Find the probability that the proportion of the defective screws $\bar{p}$ in the sample is between 0.01 and 0.03? [5mks]
- (b) Sixteen oil tins are taken at random from an automatic filling machine. The mean weight of the tins is 14.2 kg, with a standard deviation of 0.40 kg. Can we conclude that the filling machine is wasting oil by filling more than the intended weight of 14 kg, at a significance level of 5%? [5mks]
- (c) Five hundred ball bearings have a mean weight of 5.02 oz and a standard deviation of 0.30 oz. Find the probability that a random sample of 100 ball bearings chosen from this group will have a combined weight of more than 510 oz. [4mks]
- (d) The mean weekly sale of soap bars in departmental stores was 146.3 bars per store, after an advertisement campaign, the mean weekly sale in 22 stores for a typical week increased to 153.7 and shown a standard deviation of 17.2. Was the advertisement campaign successful at 1% significance level? [6mks]

QUESTION FIVE (20mks)

- (a) (i) Let T follow a t -distribution with $r = 8$ df. What is the probability that the absolute value of T is less than 2.306? [5mks]
- (ii) A company did a survey of 20 people who used its product. The mean average age of the sample was 22.4 years and the unbiased standard deviation was 3.1 years. The company now wants to advertise in a magazine that has a primary readership of people who are aged between 18 and 24. What percent of its potential customers are in this age bracket? [5mks]

- (b) In the past, the standard deviation of weights of certain 40 ounce packages filled by a machine was 0.25 ounces. A random sample of 20 packages showed a standard deviation of 0.32 ounces. Is the apparent increase in the variability significant at 0.05 level of significance? [5mks]
- (c) An instructor has two classes, A and B , in a particular subject. Class A has 16 students while class B has 25 students. On the same examination, although there was no significant difference in mean grades, class A had a standard deviation of 9 while class B had a standard deviation of 12. Can we conclude at the 0.01 level of significance that the variability of class B is greater than that of A ? [5mks]